Prepare Deployment

Slide 1 — Prepare Deployment

Prepare Deployment

- Deployment is the process of preparing an application for a production environment.
- Ensure the application is stable, secure, and ready for users.
- Main preparation steps:
  - Install production-ready dependencies
  - Configure environment variables
  - Create deployment files
  - Test the application before release

Slide 2 — Install Production-Ready Dependencies

Production Dependencies

- Install only the packages required to run the application.
- Remove unnecessary development or testing packages.
- Use fixed or compatible package versions.
- Verify that all dependencies work correctly in the production environment.

Example:

```
pip install -r requirements.txt
```

Slide 3 — Create requirements.txt

Python Dependency Management

- requirements.txt contains the Python packages required by the application.
- It helps ensure the same dependencies are installed during deployment.
- Generate it using:

```
pip freeze > requirements.txt
```

Example:

```
Flask==3.0.0
gunicorn==21.2.0
requests==2.31.0
```

Slide 4 — Set Up a Procfile

Procfile Configuration

- A Procfile defines the command used to start the application.
- It is commonly used by platforms such as Heroku and similar deployment services.
- Example for a Flask application:

```
web: gunicorn app:app
```

- web specifies the process type.
- gunicorn runs the application server.
- app:app refers to the Python file and application object.

lide 5 — Create a Dockerfile

Container-Based Deployment

- A Dockerfile contains instructions for creating a Docker image.
- It packages the application and its dependencies together.
- Example:

```
FROM python:3.11
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .
CMD ["gunicorn", "app:app"]
```

Advantages:

- Consistent environment
- Easy deployment
- Portable across systems

Slide 6 — Final Deployment Checklist

Before Deployment

- □ Install production-ready dependencies
- □ Verify `requirements.txt`
- □ Configure `Procfile` if required
- □ Create and test `Dockerfile` if using Docker
- □ Set environment variables securely
- □ Test the application
- □ Check logs and error handling